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Episode 35 · October 23, 2026 · 5 papers

Learning Dynamic Games for Cooperative Iterative Tasks

SKIM

Mark Pustilnik, Francesco Borrelli

Borrelli's group extends learning model predictive control from one robot to several players in a game. Agents reuse trajectories from earlier runs of the same task to build a shared safe set. Each player's cost then never gets worse from one run to the next.

PROBLEM Learning model predictive control gives safety and improvement guarantees when a single agent repeats a task. No comparable guarantee existed when several coupled agents with their own objectives and shared constraints repeat a task together and must each be willing to accept the new joint plan.

METHOD Feasible joint trajectories from past runs form a sampled joint safe set, and each player gets its own learned cost-to-go function. At each step the agents jointly choose a stored terminal state and a generalized Nash equilibrium that reaches it. The new plan is adopted only if it passes an agreed acceptance rule, which can encode a no-worse-off condition for each player and a selection criterion such as weighted cost or Nash bargaining.

RESULTS The abstract gives no numbers, scenarios or baselines. The stated contributions are theoretical: a proof of recursive feasibility, and conditions under which each player's cost does not increase across iterations.

CAVEAT Everything depends on the task repeating with the same start and the same players, so that stored trajectories stay valid. Merges and lane changes on open roads are one-shot encounters with changing partners, so the safe set may never fill in. The scheme also assumes the agents jointly pick the equilibrium and terminal state using full shared information. That makes it a centralized or perfect-communication design that says nothing about lost or delayed messages or partial participation. Solving generalized Nash equilibria for nonconvex games in real time is also hard, and the paper likely shows no scaling beyond a few agents.

WHY IT MATTERS It sits next to your work, not inside it. The cooperative acceptance rule with per-player non-degradation is effectively a proposal-and-acceptance scheme with a formal guarantee, and the Nash bargaining selection gives you a principled way to decide who yields. Those two pieces are worth borrowing for game-theoretic right-of-way negotiation. But the repeated-task setting and the assumption of perfect joint selection put it far from the messaging, degradation and open-road maneuver questions at the core of your research.

arXiv:2610.28901v1 PDF incremental

Machine Learning-Enabled Wireless Power Transfer: A Task-Oriented Review of Structural Optimization, State and Parameter Estimation, and Adaptive Control

SKIM

Denghao Luo et al.

This is a review of how machine learning gets used in wireless charging systems, sorted into coil design, estimating coupling and load, and adaptive control, and it has nothing to do with vehicles negotiating maneuvers.

PROBLEM Wireless power transfer systems lose efficiency and safety margins when coil alignment, coupling, load or operating conditions shift. The literature applying machine learning to this has been scattered across design, estimation and control without a common task-based way to organize it.

METHOD The authors survey prior work and group it into three task domains: structural design and optimization, parameter and state estimation, and control and scheduling. They compare supervised, reinforcement and hybrid physics-informed learning on qualitative axes, among them data needs, reliance on physical models, compute cost, accuracy and online response. No new algorithm or system is proposed.

RESULTS The abstract gives no numbers, benchmarks or quantitative comparisons. The evidence is a qualitative synthesis of earlier work, ending in a list of deployment barriers such as biased datasets, weak out-of-distribution generalization and the lack of standard benchmarks.

CAVEAT As a review, it shows nothing new, and the comparison axes look descriptive rather than measured. A reviewer would ask whether the method tradeoffs are backed by side-by-side results on shared test cases, or just compiled from papers that used incompatible setups. The paper itself notes there are no standardized benchmarks.

WHY IT MATTERS It is outside your area. It covers charging hardware and power electronics control, not negotiation, message design, platooning or joint trajectory planning between vehicles. Its one touch point is a passing mention of distributed coordination as future work, probably about scheduling chargers rather than vehicles agreeing on maneuvers, so you can safely skip it.

arXiv:2610.30101v1 PDF incremental

A Plantwide Simulation Library for Control-Oriented Design: Analytical and Data-driven Use Cases

SKIM

Kyle Territo, Luis A. Briceno-Mena, Jose A. Romagnoli

A chemical-process simulation library that stores each study's configuration as hashed plain data, demonstrated on one reactor, separator and recycle plant, where the linearized damping ratio ranks heat-exchanger networks correctly but is a poor predictor of which operating runs will oscillate.

PROBLEM Plantwide control structures and operability indices are usually evaluated on fixed simulated flowsheets, and the setup is rarely reported in a form another group could reproduce. It is also unclear whether a cheap linear index like the closed-loop damping ratio actually predicts nonlinear closed-loop behaviour.

METHOD They built plantbench, a library where plant design, controller tuning and instrumentation are each declared as hashed plain-data configurations, so any study can be identified and rerun exactly. As a case study, they ran a heuristic plantwide control design on a heat-integrated plant with an open-loop unstable reactor. They ranked four heat-exchanger networks by the least damping ratio of the linearized closed loop, then checked that ranking against 2016 closed-loop simulations that varied the network, the tuning, the analyzer deadtime and the upsets.

RESULTS The damping ratio orders the networks the same way the simulations do. The network that recovers the most heat scores 0.013 against 0.157 for the selected one, and it cycles when throughput is increased. As an oscillation detector, though, the damping ratio is weak. A random forest trained on the first 100 minutes after an upset catches 60 of 82 limit cycles, and at that same recall a damping-ratio threshold raises 173 false alarms where the forest raises 5. The integral time of the temperature loop, which the damping ratio barely responds to, partly decides whether the plant holds its operating point.

CAVEAT All the evidence comes from one simulated flowsheet with four candidate networks, so neither the reproducibility claim nor the finding that the damping ratio is a weak detector is tested on a second plant. The random forest comparison is also uneven. The forest sees 100 minutes of nonlinear trajectory data, while the damping ratio is a single linear number computed before any run, so its losing says more about the index than about the forest being clever.

WHY IT MATTERS This sits outside your area. It is about chemical process control, with no vehicles, no communication and no negotiation between agents. The one transferable lesson is a methodological one: a linear stability margin can rank designs correctly yet still miss which operating conditions produce limit cycles. That echoes the gap between linear string stability analysis and nonlinear platoon behaviour under deadtime, but you can take that point from the abstract alone.

arXiv:2610.29953v1 PDF incremental

Prefix Visibility Ratio: Quantifying Partial Observability in Carpet Bombing Detection

SKIM

Farid Naït-Abdesselam et al.

A DDoS detection paper that argues carpet bombing detectors should report what fraction of the attacked hosts they can actually see, and shows in Mininet that a detector which reweights its evidence by that fraction holds up when visibility is low.

PROBLEM Carpet bombing spreads attack traffic thinly across many addresses in a prefix, but operators usually have per-host telemetry for only some of those addresses. Existing evaluations report monitoring coverage or address utilization, and neither captures how many of the attacked destinations are actually observable.

METHOD They define the Prefix Visibility Ratio as the share of attacked destinations that have host-level telemetry available. Their detector, VANTA, computes separate anomaly scores per host and per prefix. It estimates that visibility ratio online by comparing router-side destination evidence against the list of monitored hosts, then shifts weight between the host and prefix scores according to the estimated visibility and how confident the estimate is.

RESULTS Across 2,160 Mininet runs they report 99.18 percent precision, 93.46 percent recall, a 96.24 percent F1 score and a 1.54 percent false positive rate. In a separate study at configured visibility of ten percent or less, they report 95.89 percent pooled F1 and 92.39 percent recall against reimplemented reference detectors, with the largest gap at one percent visibility.

CAVEAT Everything runs in Mininet with synthetic attacks and visibility levels the authors set themselves, and there are no real ISP traces. The baselines are the authors' own feature-level adaptations of prior systems, not the originals, and the abstract admits they do not reproduce those systems' real objectives. That makes the comparative advantage hard to trust, and the abstract gives no ablation separating the gain from visibility-aware fusion from the gain of simply having a prefix-level score.

WHY IT MATTERS This is off topic for you. It is network security and DDoS detection, with no vehicles, no negotiation and no coordination protocol. The only loose parallel is the idea of estimating at run time how much of the relevant population you can observe and weighting your decision by that estimate. That echoes coordination under partial participation, but your own literature on mixed penetration rates covers it more directly.

arXiv:2610.29802v1 PDF incremental

Multimodal Interaction Estimation for Contact-Aware Adaptive Impedance Control

SKIM

Mateusz Jaszczuk, Paris Perdikaris, Nadia Figueroa

A transformer fuses force history and point clouds to tell an unknown payload apart from a human push, so a Franka arm knows whether to compensate the load or yield compliantly.

PROBLEM A force sensor alone cannot tell a carried load from a person pushing on the arm, and vision alone fails under occlusion. That ambiguity makes it hard for a manipulator to be both precise and safely compliant around people.

METHOD They train a transformer classifier on force-torque history plus a 3D point cloud, using modality dropout so it still works when one input is missing, and fine-tune it from simulation to hardware. The predicted interaction type gates two things. One is an online system identification step that compensates for the payload only when a payload is detected. The other is a variable-impedance controller that switches between adaptive and reactive modes, and the estimated interaction force is passed on to downstream intent estimation.

RESULTS The classifier reaches 91.89 percent accuracy and stays above 90 percent when either modality is fully occluded. The control results are qualitative demonstrations: single-arm perturbation tests, use alongside an imitation-learning policy, intent-driven target switching, and a zero-shot bimanual box lift on two Franka arms. The abstract compares against no baselines.

CAVEAT Classification accuracy is the only number given, and it says nothing about closed-loop tracking error, force overshoot, or how often a misclassification makes the arm stiffen against a person. Modality dropout gives robustness to a missing input, but the abstract does not test a corrupted or misleading input. The zero-shot bimanual result covers a single box-lifting task, which is thin evidence for general closed-chain behaviour.

WHY IT MATTERS This is outside your area. It covers one robot's physical contact with humans and payloads, with no inter-agent communication, negotiation, or vehicle coordination. The only loose link is the idea of inferring intent from interaction forces, which is the inference-only approach your work is trying to replace with explicit message exchange.

arXiv:2610.24235v1 PDF incremental